

Stage 11 — standout analysis modules (each a standalone run_*.py, run in parallel as an independent Nextflow task)

Evolution & age	Regulation	Structure & engineering	Comparative & mobilization	Expression & benchmark
Subfamily phylogenetics <i>run_phylo_analysis.py</i>	Motif gain / turnover <i>run_motif_gain.py</i>	LTR structural annotation <i>run_ltr_struct.py</i>	Orthologous-insertion calling <i>run_ortholog_insertion.py</i>	Differential expression (clusters) <i>te_expression.py</i>
Repeat landscape / divergence <i>run_divergence.py</i>	Antisense / bidirectional promoter <i>run_antisense_promoter.py</i>	Protein structure (ColabFold) <i>run_fold_prediction.py</i>	Multi-assembly liftover <i>run_multiassembly_liftover.py</i>	Benchmark vs manual workflow <i>run_benchmark.py</i>
Automatic subfamily resolution <i>run_subfamily.py</i>	CTCF sites & TAD boundaries <i>run_ctcf_tad.py</i>	Allele-aware gRNA off-target <i>run_grna_offtarget.py</i>	3' transduction lineages <i>run_transduction.py</i>	
Consensus distance <i>plot_consensus_distance.py</i>	Epigenetic / regulatory overlay <i>run_epigenetic_overlay.py</i>			

All modules consume the clustered loci + consensus from the core pipeline and write their own figures and measured-value macros.